

Sheet1

Category	Component	Exact part / specification	Status
Controller	Microcontroller	ESP32-DevKitC V4 with ESP32-WROOM-32D	Have
	USB cable	Micro-USB data cable for programming and ESP32 power	Have
Sensors	Temperature and humidity	DHT11 sensor module	Have
	Light	Light Sensor Module	Have
Irrigation	Reservoir water level	Solu SL067	Have
	Soil sensor and pump	M5Stack Watering Unit — capacitive soil-moisture sensor + 5V pump	Have
	Water tubing	Tubing supplied with the M5Stack unit	Included
	Water reservoir	Small container for irrigation water	Needed
Display	Local display	0.96" SSD1306 OLED, 128 × 64, I ² C	Have
Power and wiring	Pump power supply	Regulated 5V 2.5A DC supply	Have
	Power connector	Matching female DC barrel-jack to screw-terminal adapter	Have
	Breadboard and wires	Solderless breadboard + jumper wires	Have
Plant setup	Plant	One small live plant	Needed
	Pot and tray	Pot with drainage holes + drip tray	Needed
	Soil	Potting soil suitable for the plant	Needed